

MAUVE-MUSE: When Metallicity Follows or Fights Star Formation

Virgo spirals show a mass-dependent
sign flip in the Z_{gas} - ΣSFR relation.

LOW Σ^* / outer disc

gas supply / dilution

$\uparrow \Sigma\text{SFR}$, $\downarrow Z_{\text{gas}}$

FLIP SIGN

near $\log \Sigma^*$

$\approx 7.5-8.0$

HIGH Σ^* / inner disc

star formation / enrichment

$\uparrow \Sigma\text{SFR}$, $\uparrow Z_{\text{gas}}$

Main takeaway

The same galaxy disc can switch from gas-supply-dominated regulation in the outer parts to enrichment-dominated regulation in the inner parts.

